

SIRJAN SINGH

425-240-5581 | sirjan1@cs.washington.edu | [linkedin.com/in/sirjan1](https://www.linkedin.com/in/sirjan1) | github.com/sirjanpreet

EDUCATION

University of Washington

Exp. Dec 2027

Bachelor of Science in Computer Science

Seattle, WA

- **GPA:** 3.8/4.0
- **Relevant Coursework:** Operating Systems, Distributed Systems, Machine Learning, Deep Learning for Computer Vision, Compilers, Concurrency, Networks, Natural Language Processing, Algorithms

TECHNICAL SKILLS

Systems & Infra: Linux Kernel, CUDA, Distributed Systems, Concurrency, Kubernetes, Docker, AWS (EC2/S3)

Languages: C, C++, Python, Rust, x86 Assembly, Bash

AI & ML: PyTorch, Inference Pipelines, RAG, Model Serving, Agentic Workflows

Tools: Git, GDB, Valgrind, Wireshark

EXPERIENCE

Amazon Web Services (AWS)

Fall 2026

Incoming Software Development Engineer Intern – Billing

Seattle, WA

- Anticipated to engineer concurrent, fault-tolerant components for AWS Billing's distributed architecture, processing high-throughput telemetry at scale

Expedia Group

Summer 2026

Incoming Machine Learning Engineer Intern – Book to Trip

Seattle, WA

Ericsson

June 2025 – Sept 2025

Software Engineer Intern | Python, Linux, Kubernetes

Bellevue, WA

- Developed core telemetry processing modules in Python, optimizing high-throughput data streams across **48 distributed nodes** and reducing end-to-end pipeline latency by 5%
- Engineered an automated ETL pipeline to aggregate and analyze network packet loss metrics, eliminating 10 hours/week of manual reporting and enabling real-time observability
- Orchestrated fault-tolerant deployments across 4 Kubernetes nodes, maintaining **99.7% uptime** and reducing deployment failures

NOISE Lab @ UW (Security & Privacy Research)

Sept 2024 – Mar 2025

Research Assistant | C/C++, Machine Learning, Bash

Seattle, WA

- Developed high-throughput C/C++ simulation scripts generating **4 million** synthetic network traffic data points to train machine learning-based anomaly detection models
- Implemented algorithmic defenses to mitigate real-time traffic analysis vulnerabilities, optimizing execution to maintain **sub-millisecond** latency overhead

PROJECTS

Green Thread Library | C++, POSIX, x86 Assembly

- Architected a C++ user-level threading library with a cooperative scheduler, bypassing OS kernel trap overhead and achieving **10% faster context switching** vs. native pthreads
- Implemented mutexes and condition variables to synchronize state across **10,000 concurrent lightweight tasks** with zero data races under stress testing
- Optimized async context switching to reduce switch latency by **33 μs**, targeting high-throughput use cases including HFT systems and AI inference pipelines

Thread Pool Work Stealing | Rust, Concurrency, Systems Architecture

- Developed a scalable Rust thread pool to dynamically dispatch concurrent tasks across **4 CPU cores**, achieving **90% CPU utilization** under peak load
- Programmed a work-stealing scheduler using double-ended queues to rebalance computational workloads across threads, reducing idle time by 14%

Jane Street LLM Reasoning Benchmark | Python, LLM Evaluation

- Engineered an automated benchmarking pipeline to evaluate 6 LLM reasoning models against **135 complex logic puzzles**, surfacing cross-model performance gaps at scale
- Constructed a data extraction architecture using BeautifulSoup to scrape and structure ground-truth solutions from web archives, building a clean evaluation dataset end-to-end